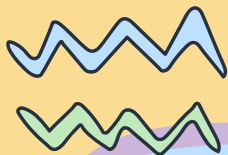


BINARY NUMBERS — HOW COMPUTERS COUNT

Fun with AI & Digital for Grade 3!





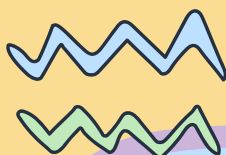
Let's Start Counting!



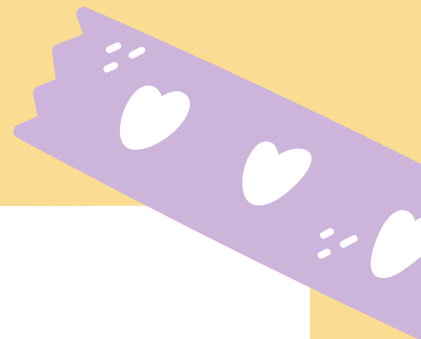
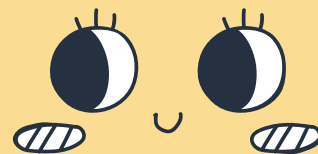
How do you count? What numbers do you use?

Think about it! When you count your toys, your friends, or your snacks — what numbers come to mind? Let's explore together!





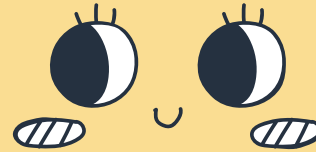
Counting with Numbers We Know



We use 10 numbers every day: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. These are called decimal numbers! When we count apples from 1 to 10, we use these same digits. Let's see how they work together!



What is Binary?



Computers use a different counting system called binary. Binary uses only two numbers: 0 and 1. That's it! While we count with ten digits, computers keep it simple with just two.



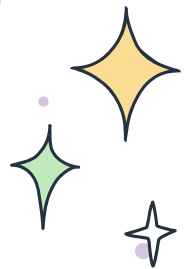
Binary Digits — Bits!

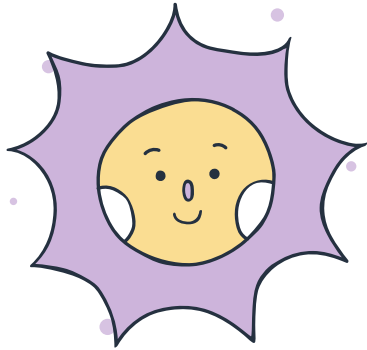
1 0 1

A 0 or 1 is called a bit. Bits are the smallest pieces of information a computer can use!

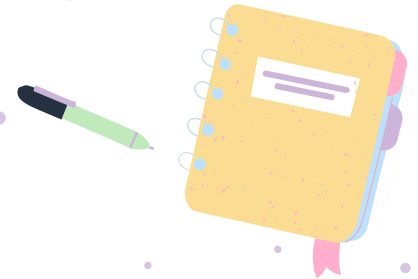
Think of bits like tiny switches — they can only be ON (1) or OFF (0). When we put bits together, we can make bigger numbers!

Fun Fact: The word "bit" comes from "binary digit"!





How Binary Counting Works — Step 1



Let's count from 0 to 3 in binary!

0 = 00 1 = 01

2 = 10 3 = 11

Place values go from right to left: 1s place, then
2s place!

0.122
05

10.01.1190
00

99.299
05

20.10
05

0



2



14



12

1



2



4



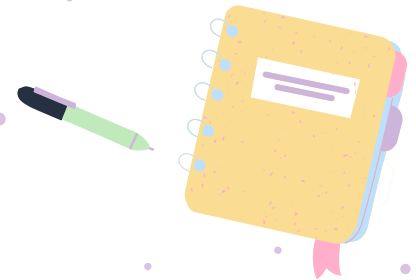
11

011





How Binary Counting Works — Step 2



Now let's count from 0 to 7 in binary!

0=000 1=001 2=010 3=011

4=100 5=101 6=110 7=111

Each place is worth double the one before it!

Decimal Numbers:

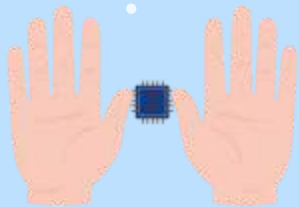
02 - 046 Decimal /number/7	03 - 030 Deciay/numberis,6	014=111 Docacy/numberisn	02 - 111 Binacy /numberiers
021-1900	507=500	013=6300	013=3900
004= 500	0828500	02511900	01114611
005=1655	000=800	00114831	00120011
004=1845	0044555	00414481	01333841



Decimal uses 10 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

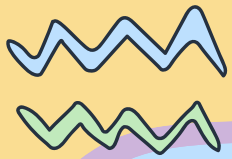
This is how we count every day! We have 10 fingers, so we use 10 numbers. Easy to remember!

Binary uses only 2 digits: 0 and 1. That's it! Just two numbers to count everything. It's like having only two fingers to count with — on or off, yes or no!

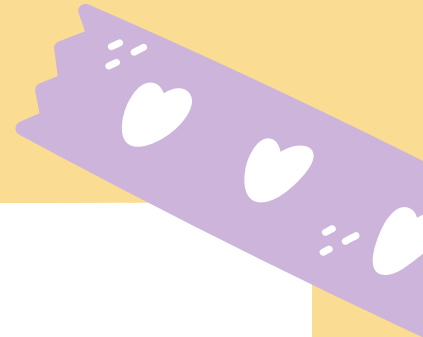
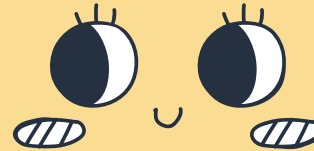


Decimal vs Binary — What's the Difference?





Why Do Computers Use Binary?



Computers use binary because it's easy to tell ON or OFF. Electric signals inside computers are either on (1) or off (0). This makes binary the perfect language for machines to understand and work with!



Binary Counting Game



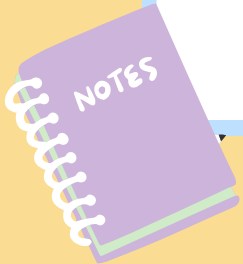
What number is this in binary?

1 0 1

Write your answer below!



Let's play a fun game! Look at the binary number above. Each bit has a place value: 4, 2, and 1 from left to right. Add up the values where you see a 1. What decimal number do you get? Try to count the place values!



Converting Binary to Decimal



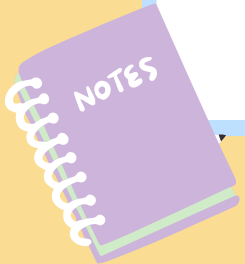
Example: 101 in binary

$$1 \times 4 + 0 \times 2 + 1 \times 1 = 5$$

So 101 in binary = 5!

$$1 \times 4 + 0 \times 2 + 1 \times 1 = 5$$

Follow these easy steps to change binary to decimal! Each place has a value: 1, 2, 4, 8 (doubling from right to left). Multiply each bit by its place value, then add them all together!



Fun Binary Codes Around Us



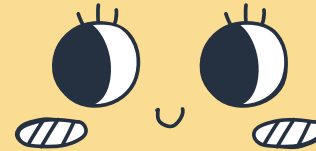
Computers use binary to
save files, run programs,
and show pictures on
screen!

Robots follow binary
instructions to move,
talk, and help people!

Video games use binary
to create characters,
sounds, and fun
adventures!

AI devices like
smartphones and smart
speakers use binary to
understand you!

Why Binary is Important for AI & Digital



Binary helps computers think and learn! Every time you talk to a smart speaker or play a video game, binary is working behind the scenes. AI uses binary to understand data and make decisions. Without binary, computers couldn't do all the amazing things they do!



Let's Recap!



Binary uses only two
numbers: 0 and 1.
That's all computers
need to count!

A bit is one binary digit.
It's the smallest piece of
computer information!

Binary counts by
doubling place values:
1, 2, 4, 8, and so on!

Computers use binary
to work, think, and
help us every day!

Can you write the number 6 in binary? Fill in
boxes below with 0 or 1!

?

?

?

Your Binary Challenge!



Hint: Remember, binary place values from right to
left are 1, 2, 4... What adds up to 6?

Try it and share with your class! You're becoming a
binary expert!

THANK YOU & KEEP EXPLORING!

Thank you for learning about binary
numbers! Keep counting, coding,
and exploring!

